Continuous random variables

Antonio de Leon

Probability and Statistics for Engineers Fall 2026

Monday, October 26	Densities, CDFs, and expectation
Wednesday, October 28	Exponential and normal models
Friday, October 30	Joint densities and marginals

Course support and deadlines

Ed Discussion: questions, announcements, and private messages. No Canvas Inbox.

Open Ed from Canvas All times Pacific Full contact details and meeting links

Instructor / TAs	Office hours	Optional, ungraded discussions
Antonio de Leon	Fri 3–4 p.m. (in person) GSC Room 204	—
Jeannine Romero	Tue / Thu 10–11 a.m.	01A: Fri 9:20–10:25 a.m. R Carson Acad 240 (in person)
Ehsan Mirafzali	Mon 3–5 p.m.	01B: Fri 10:40–11:45 a.m. (Zoom)
Seokjun Choi	Wed 3–4 p.m. 5:05–6:05 p.m.	01C: Wed 4–5:05 p.m. (Zoom)

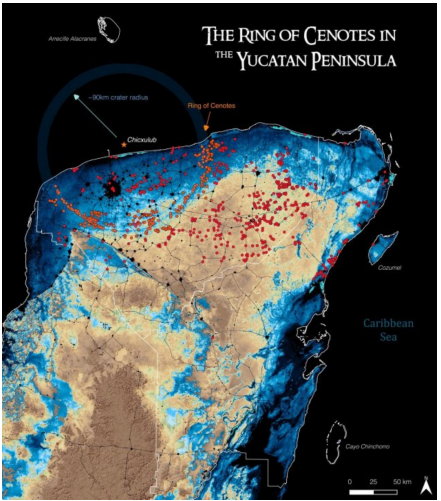
Upcoming deadlines HW and quizzes: **10 p.m. Pacific**

HW 3	**Tue, Oct 27**	Gradescope
Quiz 5	**Sun, Nov 1**	Canvas
HW 4	**Tue, Nov 3**	Gradescope

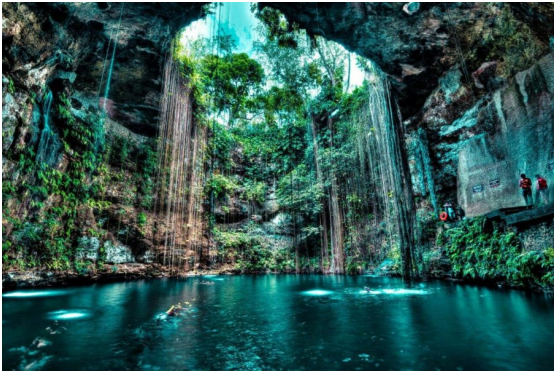
GSC: Graduate Student Commons. TA office-hour times link to Zoom.

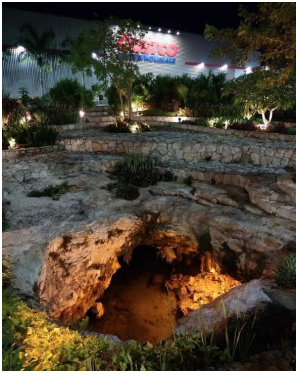
HW: 24-hour grace period. Quizzes: no grace period. Approved accommodations apply.




Arrecife Alacranes
THE RING OF CENOTES IN THE YUCATAN PENINSULA
~90km crater radius
Ring of Cenotes
Chicxulub
Cozumel
Caribbean Sea
Cayo Chinchorro
0 25 50 km

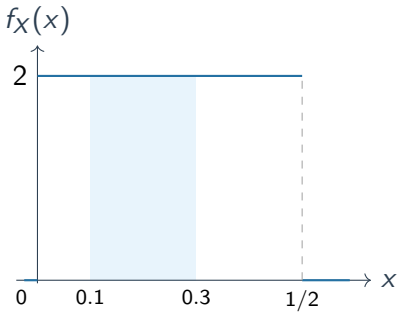






Probability from an interval

Assume X has density 2 on $(0, 1/2)$ and zero elsewhere.



Total probability
Height $\times$ width: $2(1/2) = 1$.

Shaded interval
Find $P(0.1 < X < 0.3)$.

Probability is area under the density. A density value may exceed one.
If X is in seconds, f_X has units seconds^{-1}.

Conditions for a density

A law with density f_X

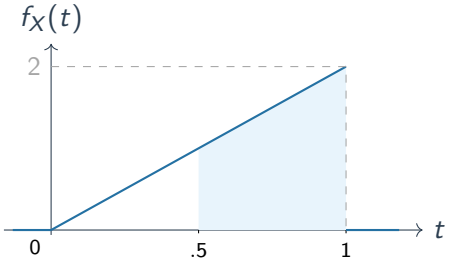
$$f_X(x) \geq 0, \qquad \int_{-\infty}^{\infty} f_X(x)\,\mathrm{d}x = 1, \qquad P(a < X \leq b) = \int_a^b f_X(x)\,\mathrm{d}x.$$

A point and an interval. What does the integral assign to $\{c\}$?

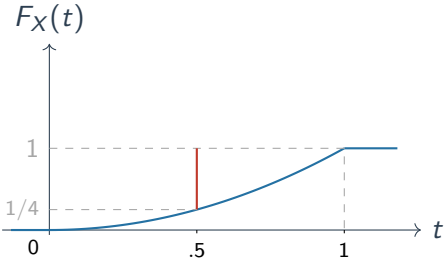
The density may have jumps. Values at isolated points do not change integrals.

Here we study laws with densities, also called absolutely continuous laws.

Density and cumulative probability



Shaded area: $P(1/2 < X \leq 1)$.



Matching change: $F_X(1) - F_X(1/2)$.

Derive the relation from $F_X(t) = P(X \leq t)$.

At a continuity point t of f_X, $F_X'(t) = f_X(t)$. The CDF is continuous.

Uniform probabilities

Let $X \sim \text{Unif}(a, b)$ with $a < b$. The density is constant on (a, b).

$$1 = c(b - a), \qquad f_X(x) = \frac{1}{b - a} I_{\{a<x<b\}}.$$

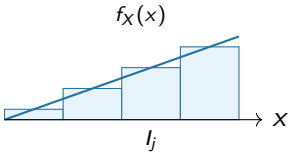
The indicator is 1 inside (a, b) and 0 outside. Endpoint density values do not change probabilities.

Find the CDF on all of $\mathbb{R}$. First locate t relative to a and b.

For $a \leq c < d \leq b$, the interval probability is $(d - c)/(b - a)$.

Expectation with a density

For bounded X, split its range into bins I_j and replace values in bin j by x_j.



Finer bins improve the approximation.

The finite rule gives

$$\mathbb{E}[g(X_\Delta)] = \sum_j g(x_j) \int_{I_j} f_X(x)\,\mathrm{d}x.$$

Let the largest bin width tend to zero.

Average the function values using probability weights. Use $g(x) = x$ or x^2.

General finite-expectation formula: require $\int |g(x)| f_X(x)\,\mathrm{d}x < \infty$.

Uniform mean and variance

Start with $U \sim \text{Unif}(0, 1)$.

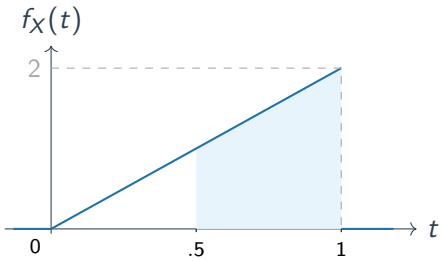
$$\mathbb{E}[U] = \int_0^1 u \, \mathrm{d}u, \qquad \mathbb{E}[U^2] = \int_0^1 u^2 \, \mathrm{d}u.$$

For $X = a + (b - a)U$, the uniform CDF gives $X \sim \text{Unif}(a, b)$.

The affine mean and variance rules carry over from Week 4.

A density that increases

Let $f_X(x) = 2x I_{\{0<x<1\}}$.



Check that the total area is one.

Normalize and obtain the full CDF.

The formula t^2 applies only between the support endpoints.

Mean and tail probability

Continue with $f_X(x) = 2x I_{\{0<x<1\}}$ and $F_X(t) = t^2$ on that interval. **Find** $P(X > 1/2)$, $\mathbb{E}[X]$, and $\mathrm{Var}(X)$.

More weight lies near one, so the mean exceeds the uniform mean $1/2$.

A polynomial density

Practice. Let $f_W(w) = cw^2 I_{\{0<w<1\}}$.

1. Find c.
2. State $F_W(t)$ for every real t.
3. Compute $\mathbb{E}[W]$.

Use the same integral rule for each part, with the appropriate integrand.

Reading a density and a CDF

For $f_X(x) = 2x I_{\{0<x<1\}}$, $F_X(t) = t^2$ on $(0, 1)$.

Quantity	Value	Interpretation
$f_X(3/4)$		
$F_X(3/4)$		
$P(X = 3/4)$		

Zero probability for one exact value does not make that value logically impossible.

Density calculations

Task	Integral or condition
Check a density	$f_X \geq 0,\ \int_{\mathbb{R}} f_X(x)\,\mathrm{d}x = 1$
Accumulate probability	$F_X(t) = \int_{-\infty}^{t} f_X(x)\,\mathrm{d}x$
Find an interval probability	$F_X(b) - F_X(a)$
Average a function	$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x) f_X(x)\,\mathrm{d}x$
Find variance	$\mathbb{E}[X^2] - (\mathbb{E}[X])^2$

Keep the support visible. Give the CDF's outside pieces. Check units and total probability.

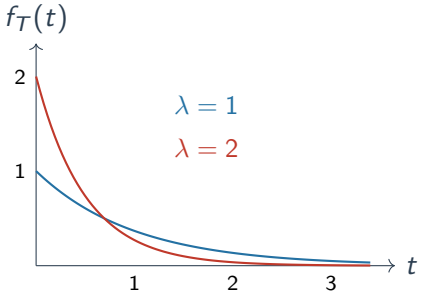
Finite expectations require absolute integrability. Variance calculations require a finite second moment.

An exponential waiting time

Assume a positive waiting time $T \sim \text{Exp}(\lambda)$, with rate $\lambda > 0$.

$$f_T(t) = \lambda e^{-\lambda t} I_{\{t>0\}}$$

Normalization



Time in minutes; rates per minute.

A larger rate places more probability near zero. Each curve has total area one.

The exponential law is a modeling assumption. Waiting times need not follow it.

CDF and survival function

For $t \geq 0$, integrate $f_T(u) = \lambda e^{-\lambda u}$ from 0 to t.

$$F_T(t) = \int_0^t \lambda e^{-\lambda u} \, \mathrm{d}u.$$

The survival function gives the probability that the wait exceeds t.
At $t = 0$, $F_T(0) = 0$ and $P(T > 0) = 1$.

Exponential mean and variance

Use integration by parts on $[0, b]$, then let $b \to \infty$.

$$\mathbb{E}[T] = \int_0^\infty t\lambda e^{-\lambda t}\,\mathrm{d}t, \qquad \mathbb{E}[T^2] = \int_0^\infty t^2\lambda e^{-\lambda t}\,\mathrm{d}t.$$

The boundary terms vanish: $b^k e^{-\lambda b} \to 0$ for $k = 1, 2$.

The remaining wait

For $s, t \geq 0$, suppose the wait has already exceeded s.



Derive $P(T > s + t \mid T > s)$ using event conditioning.

Under this model, elapsed time does not change the remaining waiting-time law.

The denominator is positive. The original T is not independent of $\{T > s\}$ in general.

Rate and time units

Assume a processing time has an exponential distribution with rate 2 per minute. **Find** the mean, variance, SD, and probability of a wait longer than half a minute.

If processing is still unfinished after three minutes, what is $P(T > 3.5 \mid T > 3)$?

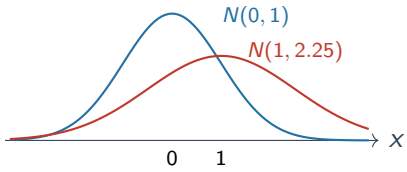
The normal density

For $\mu \in \mathbb{R}$ and $\sigma > 0$, write $X \sim N(\mu, \sigma^2)$ when

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right], \qquad x \in \mathbb{R}.$$

Standard normal functions

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \qquad \Phi(z) = \int_{-\infty}^{z} \phi(u)\,\mathrm{d}u.$$



μ locates the center. σ controls scale. The second parameter is σ^2.

We use $\int_{\mathbb{R}} e^{-z^2/2}\,\mathrm{d}z = \sqrt{2\pi}$. Its proof is in the optional appendix.

Standard-normal moments

Let Z have density ϕ. Its symmetry and decaying tails give $\mathbb{E}[Z] = 0$.

$$\phi'(z) = -z\phi(z), \qquad \mathbb{E}[Z^2] = \int_{-\infty}^{\infty} z^2 \phi(z)\,\mathrm{d}z.$$

Integrate by parts using $z^2\phi(z) = -z\phi'(z)$.

Use finite bounds first. $z\phi(z) \to 0$ at both ends, and $\int |z|\phi(z)\,\mathrm{d}z < \infty$.

Why standardization works

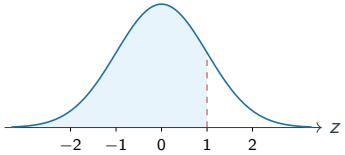
Let $X \sim N(\mu, \sigma^2)$ and set $Z = (X - \mu)/\sigma$, where $\sigma > 0$. **Find the CDF of Z.** Use $u = (x - \mu)/\sigma$ and $\mathrm{d}x = \sigma\,\mathrm{d}u$.

Thus $Z \sim N(0, 1)$. Since $X = \mu + \sigma Z$,

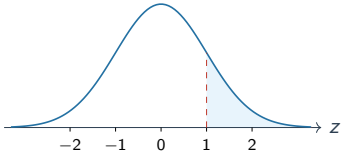
The CDF establishes the normal law. Mean zero and variance one alone do not.

Left, right, and interval probabilities

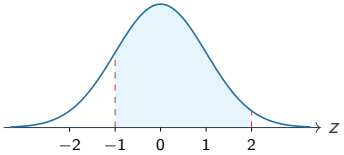
For $Z \sim N(0,1)$, $\Phi(z)$ is the area to the left of z.



$P(Z \le 1) = \Phi(1)$



$P(Z > 1) = 1 - \Phi(1)$



$P(-1 < Z < 2)$
$= \Phi(2) - \Phi(-1)$

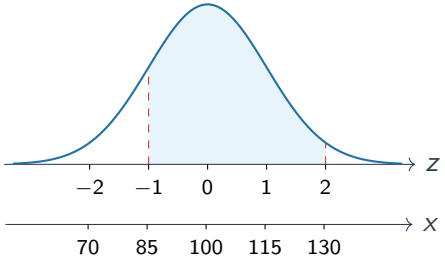
Use symmetry to express $\Phi(-z)$ through $\Phi(z)$.

z	0	1	2
$\Phi(z)$	0.500000000	0.841344746	0.977249868

The table gives numerical approximations. Shading extends into the unbounded tails.

A measurement between two limits

Assume a measurement $X \sim N(100, 225)$. Find $P(85 < X < 130)$.



First identify SD: $\sigma = \sqrt{225}$.

Original values align with standardized values.

Transform both endpoints. Centering and division by SD preserve their order.

A one-sided normal probability

Practice. Assume $W \sim N(80, 16)$. Find $P(W > 84)$.

$$\Phi(1) = 0.841344746.$$

Write the SD, standardized boundary, and correct tail.

The table reports a left-tail probability. Identify the shaded event before evaluating it.

Parameters and probabilities

	Unif(a, b)	Exp(λ)	$N(\mu, \sigma^2)$
Range	(a, b)	$(0, \infty)$	$\mathbb{R}$
Mean	$(a + b)/2$	$1/\lambda$	μ
Variance	$(b - a)^2/12$	$1/\lambda^2$	σ^2
Probability	Length ratio	$1 - e^{-\lambda t}$	$\Phi((t - \mu)/\sigma)$

λ is a rate. σ^2 is a variance. Standardization uses σ.

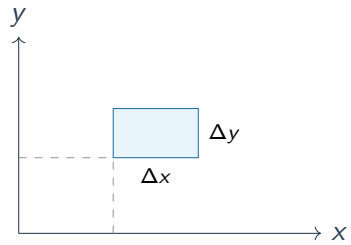
The exponential CDF formula applies for $t \geq 0$. For the uniform law, intersect the interval with (a, b).

Probability over a region

Assume (X, Y) has a joint density $f_{X,Y}$.

Probability rule

$$f_{X,Y}(x,y) \geq 0, \quad \iint_{\mathbb{R}^2} f_{X,Y}(x,y)\,dx\,dy = 1, \quad P((X,Y) \in A) = \iint_A f_{X,Y}(x,y)\,dx\,dy.$$



At a continuity point, a small cell has probability approximately

$$f_{X,Y}(x,y)\,\Delta x\,\Delta y.$$

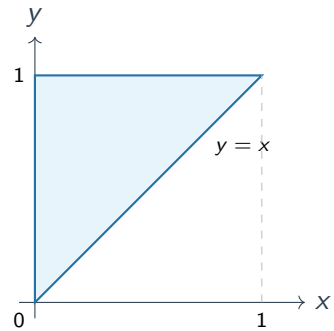
The exact probability is the integral over the cell.

The density supplies the weight; the region supplies the limits.

Continuous marginals alone do not guarantee a joint density. We assume one exists.

A uniform law on a triangle

A normalized start/finish pair has constant joint density on $S = \{(x, y) : 0 < x < y < 1\}$.



$$f_{X,Y}(x, y) = c I_{\{0<x<y<1\}}$$

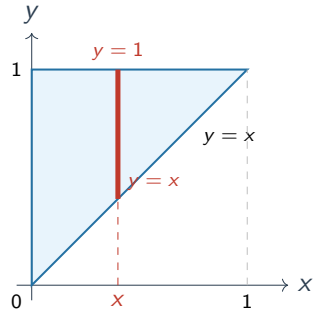
Find c**.** The total integral must be one.

Uniformity over the triangle is the model assumption.

Sequentially choosing a uniform start and a uniform finish above it gives a different law.

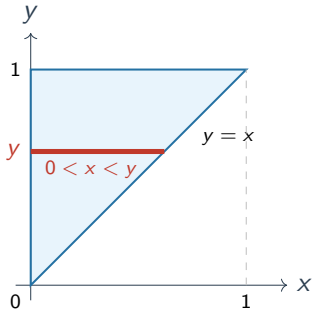
Reading the integration limits

The same region $0 < x < y < 1$ admits two slicing descriptions.



Fix x: integrate y from x to 1.

$$\int_0^1 \left[\int_x^1 f_{X,Y}(x,y)\,\mathrm{d}y \right] \mathrm{d}x$$



Fix y: integrate x from 0 to y.

$$\int_0^1 \left[\int_0^y f_{X,Y}(x,y)\,\mathrm{d}x \right] \mathrm{d}y$$

The outer variable stays fixed during the inner integration.

Deriving a marginal density

The event $a < X \leq b$ allows every compatible value of Y.

$$P(a < X \leq b) = \int_a^b \left[\int_{-\infty}^{\infty} f_{X,Y}(x, y) \, \mathrm{d}y \right] \mathrm{d}x.$$

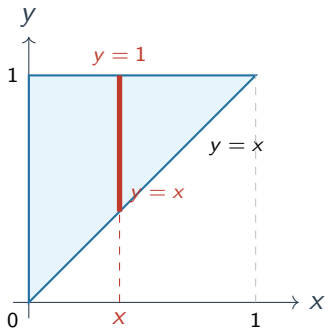
Identify the density of X. Check nonnegativity and total integral.

Integrating a slice gives a marginal density. Integrating over an event region gives probability.

Nonnegative integrands justify changing integration order here. Independence is unnecessary.

The marginal density of the start

For $f_{X,Y}(x,y) = 2I_{\{0<x<y<1\}}$, fix x and integrate over every compatible y.

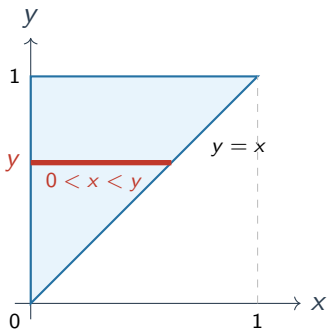


The projection onto the x-axis is $(0,1)$.

Find $f_X(x)$, including outside the projection.

The marginal density of the finish

Fix y. The compatible start coordinates satisfy $0 < x < y$.

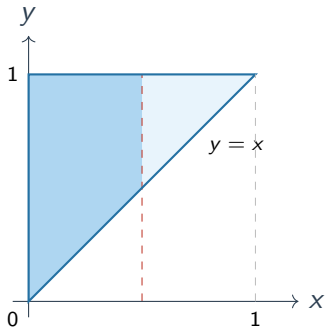


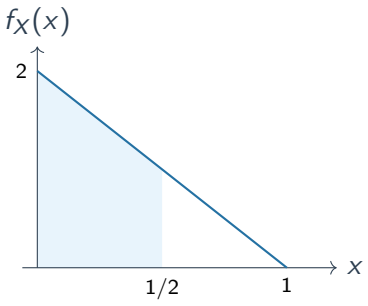
Find $f_Y(y)$ and check its integral.

The two marginals describe different coordinates of the same joint model.

A probability from a marginal

Find $P(X < 1/2)$ in the triangular model.

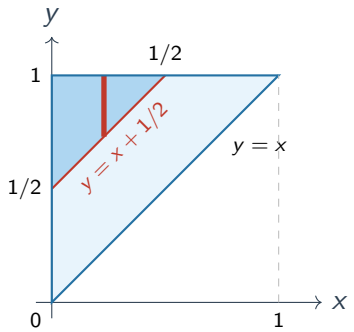




The joint-region calculation and the marginal calculation describe the same event.

A duration longer than one half

Find $P(Y - X > 1/2)$. Intersect the duration condition with $0 < x < y < 1$.

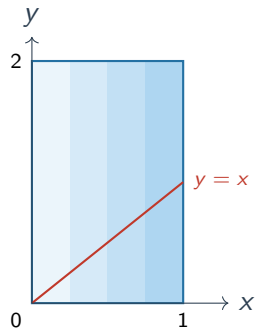


Read the limits from the shaded region.

Other order: $\int_{1/2}^{1} \int_{0}^{y-1/2} 2 \, dx \, dy$. Both integrals cover the same region.

A nonuniform joint density

Practice. Let $f_{X,Y}(x, y) = x I_{\{0<x<1,\ 0<y<2\}}$.



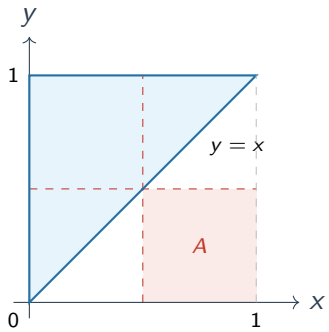
Darker bands indicate larger density.

The model is normalized: $\int_0^1 \int_0^2 x \, \mathrm{d}y \, \mathrm{d}x = 1$.

Shade $Y < X$, then calculate its probability.

Dependence from impossible combinations

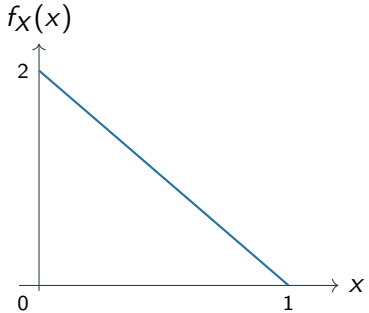
Return to the triangular model. Consider $A = \{X > 1/2,\ Y < 1/2\}$.

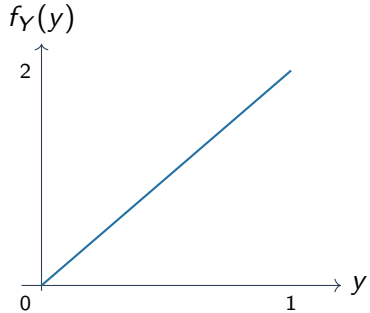


Compare the joint-event probability with the product of marginal probabilities.

Independence would require these event probabilities to factor.

Joint and marginal densities





$f_X(x) = 2(1-x)I_{\{0<x<1\}}.$

$f_Y(y) = 2yI_{\{0<y<1\}}.$

$$f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x,y)\,\mathrm{d}y, \qquad f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x,y)\,\mathrm{d}x.$$

Integrate over the compatible coordinate. The support diagram determines the limits.

Next meeting: conditional densities and fixed conditional means.

Gaussian normalization

Optional proof. Requires double integration and the polar area element.

Put $I = \int_{\mathbb{R}} e^{-z^2/2}\,\mathrm{d}z$. We seek the normalizing constant.

$$I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)/2}\,\mathrm{d}x\,\mathrm{d}y.$$

Use $x = r\cos\theta$, $y = r\sin\theta$, and area element $r\,\mathrm{d}r\,\mathrm{d}\theta$.

$I < \infty$: for $|z| \geq 1$, $e^{-z^2/2} \leq e^{-|z|/2}$. Polar area is radius $\times$ radial width $\times$ angular width.

A normal percentile

Optional application. A percentile is a threshold in a specified distribution.

For $X \sim N(\mu, \sigma^2)$ and $0 < p < 1$, find q_p such that $P(X \leq q_p) = p$.

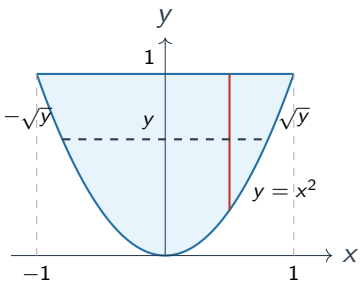
For $N(100, 225)$, use $\Phi^{-1}(0.975) \approx 1.96$.

About 97.5% of this model's probability lies below the threshold.

A density on curved support

Optional extension. The same slicing method applies to a curved boundary.

Let $f_{X,Y}(x,y) = cx^2 y I_{\{x^2<y<1\}}$.



Find c. The x-projection is $(-1, 1)$.

Marginals on curved support

Optional extension. Here $f_{X,Y}(x,y) = (21/4)x^2 y I_{\{x^2<y<1\}}$.

Vertical slice

$-1 < x < 1$, $x^2 < y < 1$.

Horizontal slice

$0 < y < 1$, $-\sqrt{y} < x < \sqrt{y}$.

The horizontal slice uses both branches, $-\sqrt{y}$ and $\sqrt{y}$.

Reading selections

Harchol-Balter, supplied 2024 version

Monday	pp. 134–137 and 141: density/CDF, uniform model, moments.
Wednesday	pp. 138–139, 142 (exponential); pp. 170–175 (normal moments and probabilities).
Friday	pp. 153–155: joint densities and marginals. Stop before conditional densities.

DeGroot–Schervish, fourth edition

Densities and moments	pp. 101–104, 111–112, 209–210; continuous expectation statement on pp. 213–214.
Named models	p. 303 and pp. 307–308 (normal); pp. 321–322, exponential portions only.
Joint laws	pp. 120–122 and the marginalization theorem on p. 132; p. 133 for the optional curved example.

Printed pages. Read selected passages alongside lecture; whole chapters are not assigned.

Exclude surrounding MGF, gamma, conditional-density, normal-sum, and CLT material. Optional frames 36–39 extend the core.